Robust Reconstruction of Tokamak Plasma State from Sparse Diagnostics Using Continuous-Time Neural Networks

Evaluating Latent Dynamics Models (CfC/LTC) Under Sensor Corruption

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Problem Statement & Motivation

Problem. Tokamaks confine plasma using magnetic fields; nothing can physically probe the plasma itself, so its magnetic state – the 2D poloidal flux map $\psi(R, Z)$, which determines plasma shape and position – must be reconstructed from sparse external magnetic sensors (flux loops, pickup coils, coil currents).

Status quo. Deep learning models (CNN, CNN+LSTM, and recently a transformer foundation model) already do this well **on clean sensor data**. But real magnetic diagnostics degrade, drop out, or fail entirely during operation – especially relevant for next-generation reactors where neutron damage will degrade in-vessel magnetics over time.

What's missing. Sensor-corruption robustness has been rigorously measured for a **forecasting** task (predicting a future scalar quantity), where the failure was shown to be severe – but this evaluation is comparatively less studied for **reconstruction** (recovering the current spatial state), a distinct task with different input-output structure. We test this directly, and evaluate whether a continuous-time encoder is more robust to it than the standard discrete-time baseline.

Background / Problem Context

- A tokamak's magnetic field shapes the plasma into nested surfaces described by the **poloidal flux** $\psi(R, Z)$ – a 2D scalar field, not directly measurable, traditionally solved for by an iterative physics code (EFIT) from boundary magnetic measurements.
- Real diagnostics are **heterogeneous and imperfect**: flux loops, pickup coils, and coil currents sample at different rates and can be noisy, delayed, or missing (MAST diagnostics show measurable missing-data rates in practice).
- Deep learning offers a **fast, direct** sensor-to-field map, replacing EFIT's iterative solve – already demonstrated to work well on clean data (TokaMark, TokaMind).
- **Continuous-time latent dynamics** (Neural ODE, LTC, CfC) are built to handle irregular timing and missing inputs by explicit elapsed-time conditioning – already used elsewhere in tokamak research for plasma transport dynamics, but never for spatial reconstruction.

Literature Review – Overview

Four threads of prior work anchor this project (full set of $\sim$15 papers in the separate Literature Review Table):

A. Reconstruction on real tokamak data – dataset, task, and baselines

B. Robustness to missing/corrupted sensors – documented, but only for a forecasting task

C. Continuous-time models in tokamak research – applied to dynamics, never to reconstruction

D. Physics-informed / alternative reconstruction approaches

Literature Review – Thread A: Reconstruction on Real Tokamak Data

TokaMark (Rousseau et al., IBM Research + UKAEA, 2026) is a public benchmark built on 11,573 real MAST tokamak shots (FAIR-MAST), defining standardized reconstruction tasks including flux-map (Task 1-3) and plasma-boundary (Task 1-2) reconstruction from sparse magnetics, with published CNN and CNN+LSTM baselines. **This is our dataset and our baseline to beat.**

TokaMind (Boschi et al., 2026), a multimodal transformer foundation model, outperforms TokaMark's strongest baseline (v3-comparable, Table 2) and supports optional token/chunk-dropout regularization for missing-diagnostic robustness (TokaMind §5.1) – but reports only clean-data accuracy; it does not measure how much its performance degrades under controlled sensor corruption at test time.

Related real-device work – a DIII-D plasma-boundary reconstruction study from reduced diagnostics (2026), and multimodal cross-diagnostic super-resolution on DIII-D (Jalalvand et al., *Nat. Commun.*, 2025) – confirms reconstruction from sparse/partial sensors works well **in-distribution**, and that handling degraded diagnostics is an active, current research direction industry-wide.

Literature Review – Thread B: Robustness – Forecasting vs. Reconstruction

Gupta (2026) ran a systematic robustness benchmark on TokaMark: LSTM, Transformer, CNN, and XGBoost trained clean, then evaluated under six controlled sensor-corruption scenarios, for **Task 4-4 – a scalar *forecasting* task** (predicting future plasma current). Result: LSTM degrades by up to **+212% NRMSE** and its disruption-alarm detection rate collapses to **0%** under sensor failure near the critical event. Forward-fill imputation largely recovers random-dropout losses.

Important distinction. Forecasting predicts a future value from past observations; reconstruction recovers the *current* spatial state from concurrent sparse sensors – a different input-output structure. Gupta's train-clean/corrupt-at-test methodology has not, to our knowledge, been applied to the reconstruction tasks (Task 1-2/1-3), on any architecture – this evaluation gap, specific to reconstruction, is what we address.

FusionMAE (Yang et al., 2025) handles missing diagnostic channels via masked-autoencoder pretraining (HL-3, 88 channels, 96.7% masked- channel reconstruction accuracy) – another data-level fix, evaluated on signal imputation, not on 2D spatial-field reconstruction quality.

Literature Review – Thread C: Continuous-Time Models in Tokamak Research

Continuous-time models are **not new to tokamak research** – but every application found targets a different problem than ours:

- Neural ODEs have been used to model **plasma transport dynamics** (core/edge/SOL energy exchange) on DIII-D data, and to forecast scalar quantities like plasma inductance on Alcator C-Mod – 0D physics-parameter fitting and scalar forecasting, not spatial reconstruction.
- A **Continuous CNN** (Arnold, Spangher & Rea, 2023) – continuous convolution kernels, a different continuous-time mechanism from CfC/LTC – significantly outperformed discrete CNNs for disruption prediction on Alcator C-Mod (AUC 0.974 vs. 0.799), but the authors explicitly state they have **not yet tested robustness** to missing/corrupted data – flagged as their own open problem.

No application of any continuous-time architecture (Neural ODE, LTC, or CfC) to spatial plasma-state reconstruction was found. This is the precise, narrow gap this project addresses.

Literature Review – Comparison Table (Core Papers)

Paper (Year)		Task	Data	Key Result	Limitation / Relevance
Rousseau et al., TokaMark (arXiv:2602.10132v3)	2026	Flux map / boundary reconstruction, forecasting (14 tasks)	11,573 real MAST shots (FAIR-MAST)	CNN+LSTM Task 1-3: 0.159 NRMSE (random) $\rightarrow$ 0.257 (temporal) (v3, Table 3; CNN: 0.148 $\rightarrow$ 0.502)	**Our dataset and baseline**; no robustness-to-corruption evaluation
Boschi et al., TokaMind (arXiv:2602.15084)	2026	Multimodal foundation model, all TokaMark tasks	Same (FAIR-MAST)	Beats strongest baseline on Task 1-3 (FT-Base 0.122 vs CNN 0.148/C+LSTM 0.159); optional token/chunk-dropout reg. (§5.1)	No systematic test-time corruption benchmark reported
Gupta, Robustness Benchmark (arXiv:2607.11915)	2026	*Forecasting* (Task 4-4, plasma current) under 6 corruption scenarios	Same (TokaMark)	LSTM: +212% NRMSE, 0% alarm TPR under event-proximate failure	**Methodology template**; never applied to reconstruction tasks
Arnold, Spangher & Rea, Continuous CNN (arXiv:2312.01286)	2023	Disruption prediction (classification)	Alcator C-Mod	Continuous CNN AUC 0.974 vs. discrete CNN 0.799	Different continuous-time mechanism; robustness explicitly untested (their own stated limitation)
Rousseau et al. *(consortium)*, Fusion Equilibrium Challenge (arXiv:2609.01750)	2026	Flux map reconstruction from non-magnetic diagnostics	9,113 DIII-D + 2,416 MAST shots (arXiv:2609.01750)	SSIM 0.83 in-machine $\rightarrow$ 0.10 cross-machine	No temporal modeling at all; cross-machine, not corruption, robustness

Research Gap

- Reconstruction from sparse magnetic sensors already performs well **in-distribution** (TokaMark, TokaMind) – accuracy itself is comparatively well studied and is not the open problem here.
- Sensor-corruption robustness has been rigorously benchmarked, but **only for a forecasting task** (Gupta, 2026); the same train-clean/corrupt-at-test methodology is **comparatively less studied** for reconstruction tasks – to our knowledge, not yet applied there.
- Existing fixes for missing/degraded sensors are **data-level** (imputation, masked pretraining) – an **architectural** fix is limited work by comparison.
- Continuous-time models are used in tokamak research (transport dynamics, disruption prediction), but their application to **spatial state reconstruction** appears **underexplored** in the literature surveyed.

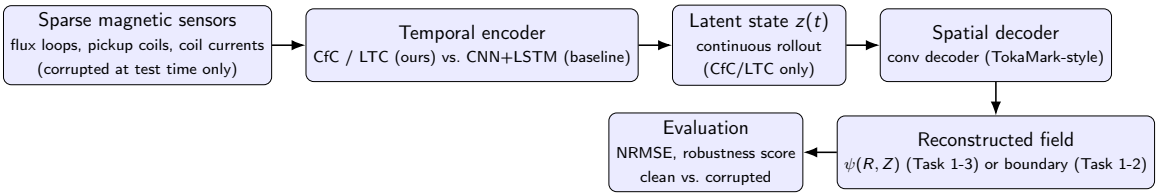
Gap, stated precisely (a limited-work claim, not "first-ever")

An **underexplored evaluation**: whether reconstruction models – discrete- or continuous-time – degrade gracefully or catastrophically under sensor corruption, measured like-for-like with imputation controlled – and whether continuous-time encoders offer a measurable robustness advantage on top of imputation. Clean reconstruction itself is the solved baseline, not the claim.

Project Objectives

1. E1 (correctness check, not a claim): reproduce CNN/CNN+LSTM on clean Task 1-3/1-2 to verify the pipeline against v3 Table 3.
2. Implement a continuous-time encoder (CfC primary; LTC as a contrasting variant) in place of the CNN+LSTM temporal backbone.
3. Define and apply a controlled sensor-corruption protocol (random dropout, single-channel failure, temporal gap) to the test set only – adapting Gupta's methodology from forecasting to reconstruction.
4. E2 (main): compare degradation of CNN+LSTM vs. CfC/LTC at equal imputation (zero-fill / forward-fill / no-imputation), using NRMSE and a robustness score – testing whether architecture adds anything on top of imputation.
5. E3 (secondary/exploratory only): temporal-split check for campaign-shift generalization, not part of the core claim.

Proposed Methodology – Architecture & Data Flow



Key differentiator vs. prior art: this is a controlled, like-for-like substitution – same task, same data, same decoder – so the CNN+LSTM-vs-CfC/LTC comparison isolates the effect of the temporal encoder, under both clean and corrupted conditions.

Proposed Methodology – Sensor Corruption Protocol & Loss

Corruption scenarios (applied to the test set only, after training on clean data – following Gupta's train-clean/corrupt-at-test design):

- **Random dropout**: randomly zero a fraction (e.g. 10% / 25% / 50%) of sensor readings across channels and timesteps
- **Single-channel failure**: one full sensor channel (e.g. a flux loop) removed for the entire input window
- **Temporal gap**: a contiguous block of timesteps blanked out (simulating an acquisition dropout)

Continuous-time formulation. The latent state evolves as a function of elapsed time, not step count:

$$\frac{dz(t)}{dt} = f_\theta\big(z(t), x(t)\big), \qquad z(t) = z(t_0) + \int_{t_0}^{t} f_\theta\big(z(\tau), x(\tau)\big)\, d\tau.$$

CfC solves this in closed form (no numerical solver), so a missing or delayed reading changes the elapsed-time input rather than silently shifting the sequence index, as it would for an LSTM.

Loss (reconstruction, matching TokaMark's own metric):

$$\mathcal{L}_{\text{recon}} = \text{NRMSE}\big(\hat{\psi}(R, Z),\, \psi(R, Z)\big)$$

Uniqueness / Novelty / Contribution

- Extends Gupta's (2026) sensor-corruption benchmarking methodology – established for a **forecasting** task – to a **spatial reconstruction** task (Task 1-2/1-3), a setting this evaluation approach appears **comparatively less studied** in.
- Applies a continuous-time encoder (CfC, contrasted with LTC) to plasma state reconstruction – continuous-time models are established elsewhere in tokamak research (transport dynamics, disruption prediction) but their use for spatial reconstruction is **limited work** by comparison.
- A **controlled, like-for-like architectural comparison** (same data, same decoder, only the temporal encoder changes) against an existing model class that has already shown promise on tokamak time series (Arnold et al.'s Continuous CNN result), evaluated here under a robustness lens that prior work did not apply.
- The contribution is framed as a **robustness evaluation and comparison**, not a claim of a new state-of-the-art architecture: the standard CNN+LSTM baseline may already be adequately robust, and that would be a valid, reportable outcome of this study.

Dataset & Experimental Setup

Data source. TokaMark / FAIR-MAST – real experimental shots from MAST. Publicly streamable, no credentials required (`s3.echo.stfc.ac.uk`); only the magnetics + currents + flux-map signals needed for Task 1-3 (9 of 39 total signal types) are streamed, not the full $\sim$2 TB archive.

Tasks. Task 1-3 (flux map $\psi(R, Z)$, primary) and Task 1-2 (plasma boundary, secondary/simpler check), both already defined with published baselines by TokaMark.

Baselines (real, published numbers to reproduce and compare against).

- CNN: 0.148 (Task 1-3, random split) / 0.502 (temporal split)
- CNN+LSTM: 0.159 (Task 1-3, random split) / 0.257 (temporal split)

Corruption protocol. Random dropout (10/25/50%), single-channel failure, temporal gap – applied to the test split only, following Gupta's clean-train / corrupt-test design (Section: Methodology).

Environment. Python 3.10, PyTorch 2.3 on a single NVIDIA A100 GPU, using `ncps` 1.0.1 for the CfC/LTC cells.

Implementation Status / Preliminary Results

- ☒ TokaMark data loader set up, magnetics+currents+flux streamed (done – verified on real shots by EDA)
- ☐ CNN/CNN+LSTM baselines reproduced (in progress – Kaggle run pending; targets CNN 0.148 / CNN+LSTM 0.159)
- ☐ CfC/LTC encoder implemented
- ☐ Corruption protocol implemented (dropout, channel failure, temporal gap)
- ☐ Preliminary clean-vs-corrupted comparison

FIGURE PLACEHOLDER

reconstructed $\psi(R, Z)$ vs. ground truth, or

NRMSE vs. corruption severity curve

Challenges / Limitations

- **Scope**: this project reconstructs one field (flux map or boundary), not the full multi-field plasma state (temperature, density, velocity together) – a real limitation, left as future work.
- **Not full reactor control**: flux/boundary is one input a real control system needs, alongside temperature, density, and disruption precursors – this project does not attempt control.
- **CfC implementation cost**: unlike LSTM, CfC is not a standard library layer; correct implementation needs care.
- **Baseline-number stability**: TokaMark's published numbers have already changed between paper revisions – our reproduced baseline must be verified against the live version, not assumed.
- **Negative-result risk**: it is possible the discrete-time baseline is already adequately robust – a valid, reportable outcome, not a failure of the project.

Work Plan & Timeline

Task	Status	Owner(s)
Literature review & gap finalization	Completed	Kalidasan
TokaMark data pipeline setup (Task 1-2/1-3)	In progress	Nishanth
Reproduce CNN / CNN+LSTM baselines	In progress	Kalidasan
Implement CfC / LTC encoder	Remaining	Nishanth
Implement corruption protocol	Remaining	Kalidasan
E1: clean-data baseline reproduction	Remaining	Nishanth
E2: robustness comparison under corruption	Remaining	Nishanth
E3: temporal-split generalization check	Remaining	Kalidasan
Result analysis & writeup	Remaining	Nishanth
Final report & presentation	Remaining	Kalidasan & Nishanth

Baseline reproduction is expected to complete within the next 1–2 days, keeping the remaining experiments on track within the mid-review timeline.

Expected Outcomes & Evaluation Plan

Metrics. NRMSE (matching TokaMark's own metric, for direct comparability); a robustness score (relative degradation from clean to corrupted, following Gupta's framing).

Experiments.

- E1 Reproduce CNN/CNN+LSTM baseline on clean data (correctness check)
- E2 Corruption sweep (dropout / channel failure / temporal gap) – CNN+LSTM vs. CfC/LTC
- E3 Temporal-split evaluation (does continuous-time also help generalization, not just robustness)

Anticipated outcome (a comparison, not a predetermined result)

On clean data, CfC/LTC is anticipated to be comparable to CNN+LSTM (E1). Under corruption (E2), the explicit elapsed-time conditioning of continuous-time encoders **may** support more graceful degradation relative to the discrete-time baseline – this is the hypothesis under test, not an assumed conclusion. Either outcome, including a negative result showing no meaningful difference, will be reported as a valid finding of this evaluation.

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